Joseph Li

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EDUCATION

University of Southern California — **Los Angeles, CA**
M.S. Computer Science (Artificial Intelligence) | GPA: 3.58/4.00 — *Sep 2024 – May 2026*
Relevant Coursework: Machine Learning, Natural Language Processing, Analysis of Algorithms, Database Systems

University of Michigan, Ann Arbor — **Ann Arbor, MI**
B.S.E. Data Science | GPA: 3.68/4.00 — *Sep 2020 – May 2024*

PROFESSIONAL EXPERIENCE

Cortex AI — *AI coding workspace and model routing platform* — **Menlo Park, CA**
AI Engineer Intern — *Jun 2026 – Aug 2026*

- Built a public token-exchange API that authenticates and meters per-customer spend on third-party models for a 2,000-seat enterprise customer
- Authored 150+ Harbor-style agentic task packages, each pairing a prompt with a synthetically seeded workspace fixture, clean-state reset, tool and time budgets, and forbidden-tool constraints, run nightly in CI as a merge gate on pass rate
- Designed the grading stack: per-task verifier scripts emitting binary reward on the artifact, an LLM-judge rubric penalizing excess tool calls, and a 22-mode failure taxonomy that isolated unbounded retry loops as the dominant class
- Designed a server-side Langfuse ingest proxy (TypeScript) capturing full tool-call trajectories per run, normalizing model IDs across 9 upstream providers and reconciling per-run token cost against the billing ledger
- Cut per-request token cost 15% on identical tasks by eliminating redundant tool calls and tightening context-window management

Yoyo Markets Inc. — *AI-native prediction markets startup* — **New York, NY (Remote)**
AI Engineer (Part-time) — *Aug 2025 – Dec 2025*

- Led a 3-person team building a generation pipeline that turns real-time social signals into candidate prediction markets, producing 50+ published markets per week versus hand-curation (5x)
- Designed a multi-agent pairwise-preference ranker (ELO over 200+ candidates per cycle) as the quality gate, choosing pairwise LLM judgments over absolute scores for calibration, lifting published market quality 30%
- Built an async Python collection service polling 150+ high-signal social accounts concurrently, expanding the signal pool 3x beyond public trend aggregators
- Shipped a Next.js/Vercel human-in-the-loop review dashboard the founding team used daily to approve generated markets, cutting turnaround from 30 minutes to under 5

RESEARCH AND PROJECTS

Weenix (A Unix-Like OS Kernel) — **USC**
C, x86, QEMU — *Fall 2025*

- Implemented processes, kernel threads, a round-robin scheduler, context switching, and kernel mutexes
- Built the virtual file system and virtual memory layers: demand paging, mmap, page-fault handling, and copy-on-write fork

University of Michigan, SOCR Lab — **Ann Arbor, MI**
Privacy-Preserving Clinical Text Obfuscation — *2023 – 2024*

- Built an NLP obfuscation pipeline using named entity recognition and differential privacy to de-identify 10,000+ clinical patient records for research use without exposing patient identity
- Benchmarked 3 pipeline variants, reaching 94% entity-redaction accuracy while keeping downstream task performance within 3% of the unmasked baseline

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, SQL
Frameworks & Infrastructure: React, Next.js, Docker, PostgreSQL, Git
AI & ML: PyTorch, NLP, Agentic Evals (Harbor), Verifier & Reward Design, LLM-as-judge, Synthetic Data Pipelines, Multi-Agent Systems, Langfuse